INSTITUTE AND FACULTY OF ACTUARIES

EXAMINATION

15 April 2025

Subject CS1 – Actuarial Statistics
Core Principles

Paper A

Time allowed: Three hours and twenty minutes

In addition to this paper you should have available the 2002 edition of the Formulae and Tables and your own electronic calculator.

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1 The marketing department of a vehicle breakdown recovery company has carried out a survey, which asked a random sample of people how many cars were in their household.

Due to a data corruption error, the frequency for the category of four cars has been lost and is denoted below by x, but the sample mean is known (to three decimal places).

Number of cars in household	*Frequency*
0	6
1	13
2	11
3	6
4	x

Sample mean: 1.667 (three decimal places).

(i) Identify which **one** of the following options gives the correct value of x:

A 6
B 5
C 4
D 3.

[2]

(ii) Determine:

(a) the mode of the sample. [1]

(b) the median of the sample. [2]

One of the company's analysts says that both the mode and median of the sample could have been determined without knowing the value of x or the sample mean by assuming that x was small.

(iii) Justify whether the analyst's statements are correct or not. [4]

[Total 9]

2 A random variable, X, taking values 1, 2, 3, …, follows a geometric distribution with unknown parameter, p, where $0 < p < 1$. It is assumed that the prior distribution of p has probability density function given as:

$$\frac{\Gamma(12)}{\Gamma(5)\Gamma(7)}\ p^4(1-p)^6$$

A random sample of X, with size 20 has a sample mean 26.

(i) State the distribution family to which the prior distribution of p belongs. [1]

(ii) Determine the posterior distribution of p together with all its parameters. [4]

[Total 5]

3 Consider the following function:

$$f(x) = \begin{cases} \frac{1}{2}\lambda e^{-\lambda x} + \frac{1}{2}\mu e^{-\mu x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

where $\lambda > 0$ and $\mu > 0$ are parameters.

(i) Determine the range of values for $\mu > 0$ for which $f(x)$ is a probability density function for any positive value of λ. [3]

Assuming that λ and μ are chosen such that $f(x)$ is a probability density function, let X be a random variable with probability density function $f(x)$.

(ii) Identify which one of the following expressions gives the expectation of X:

A $\frac{1}{2}(\lambda + \mu)$

B $\frac{1}{2}\left(\frac{1}{\lambda} + \frac{1}{\mu}\right)$

C $\frac{1}{2}\left(\frac{\lambda}{\mu}\right)$

D $\frac{1}{2}\left(\frac{\mu}{\lambda}\right)$.

[2]

(iii) Identify which one of the following expressions gives the second moment of X:

A $\left(\frac{1}{\lambda} + \frac{1}{\mu}\right)^2$

B $\frac{1}{2}\left(\frac{1}{\lambda} + \frac{1}{\mu}\right)^2$

C $\frac{1}{\lambda^2} + \frac{1}{\mu^2}$

D $\frac{1}{2}\left(\frac{1}{\lambda^2} + \frac{1}{\mu^2}\right)$.

[3]

(iv) Determine the variance of X in terms of λ and μ. [2]

[Total 10]

4 Consider the random variable X with probability density function:

$$f(x) = x\, e^{-x}, \text{ for } x > 0.$$

(i) Identify which one of the following options gives the moment generating function of X for $t < 1$:

A $M(t) = (1 - t)^{-1}$
B $M(t) = (1 - t)^{-2}$
C $M(t) = (1 - t)^{2}$
D $M(t) = (1 - t)$.

[2]

(ii) Determine the expected value and the variance of X, using your answer to part (i). [3]

(iii) Show that the cumulative distribution function of X is given by:

$$F(x) = 1 - (1 + x)e^{-x}, \text{ for } x > 0.$$

[3]

(iv) Determine the probability $P(1 \leq X \leq 2)$. [2]

[Total 10]

5 In a Generalised Linear Model (GLM) analysis, the number of claims per year (Y) on an insurance policy follows a Poisson distribution. The linear predictor is given as:

$$\ln(\mu) = \beta_0 + \beta_1 x$$

where μ is the expected number of claims and x is a policyholder's age.

(i) Explain the purpose of the link function in a GLM, also referring to the particular function used for the linear predictor above. [2]

(ii) Determine the expected number of claims, μ_1, when the policyholder's age is $x + 1$, in terms of μ and β_1. [2]

(iii) Comment, using your answer in part (ii), on how the regression coefficient β_1 differs from the slope coefficient in a linear regression model. [3]

[Total 7]

6 A climate scientist is studying the number of floods in different areas. In each area, floods are believed to occur at a constant yearly rate $\theta > 0$, which is the same in all areas. The scientist assumes that the yearly number of floods, Y, in an area follows a Poisson(θ) distribution.

The following sample data have been collected from ten areas, giving the number of floods during the past year:

Area 1	*Area 2*	*Area 3*	*Area 4*	*Area 5*	*Area 6*	*Area 7*	*Area 8*	*Area 9*	*Area 10*
0	1	0	1	5	3	1	4	1	0

You can assume that the number of floods in one area is independent of floods in other areas.

(i) Calculate the sample mean and the sample variance of this data. [1]

(ii) Explain briefly why the assumption of a Poisson distribution for Y may not be appropriate in the light of the observed data. [1]

The scientist now wants to adopt a Bayesian approach. In addition to $Y \sim$ Poisson(θ), the scientist also assumes that θ has a gamma prior distribution with parameters (a, b).

(iii) Determine the posterior distribution of θ, together with all its parameters, in terms of a and b. [4]

The scientist also wants to explore a different prior distribution and assumes that θ follows a lognormal prior distribution with parameters m and s^2.

(iv) Determine an expression for the posterior density of θ, in terms of m and s^2, up to an appropriate constant of proportionality. [4]

(v) Discuss why, from a practical point of view, the scientist may prefer to use a gamma prior distribution for θ, as opposed to a lognormal prior distribution. Your answer should refer to the form of the posterior distribution based on the choice of the prior distribution. [2]

[Total 12]

7 It is assumed that the speed of vehicles on a certain rural road can be modelled as a normally distributed random variable denoted by X. An insurer has installed a 'black box' device in each of its customers' vehicles, which records various data including speed. The insurer has used these to obtain a random sample of X denoted by $x_1, \ldots, x_n$. The sample size is $n = 64$, the sample mean is $\bar{x} = 64.2$ mph and the sample standard deviation is $s = 4.8$ mph.

[**Hint:** When calculating confidence intervals below, you can use the nearest percentage point available in statistical tables, where appropriate.]

(i) Calculate a two-sided 95% confidence interval for the expectation of X, assuming that the variance of X is known and given by $\sigma^2 = 4.8^2$. [1]

(ii) Calculate a 95% confidence interval for the expectation of X, assuming that the variance of X is unknown. [1]

(iii) Compare, and comment on, the results in parts (i) and (ii). [3]

An analyst has calculated the upper limit of a one-sided 99% confidence interval based on the sample to be 65.6 mph.

(iv) Show that the upper limit of the one-sided 99% confidence interval is 65.634 to three decimal places. [1]

The analyst has made the following statement:

'Based on a sample of 64 recorded speed measurements obtained from our black boxes, we found that the average speed on this road is less than 65.6 mph with a probability of 0.99.'

(v) Comment on the validity of the analyst's statement. [1]

The insurer wants to repeat the investigation for a similar road using the knowledge they have gained from the above analysis. The insurer wishes to take a random sample from this second road. Denote the speed on this road by Y and assume a normal distribution with expectation μ_Y and variance σ_Y^2, i.e. $Y \sim N(\mu_Y, \sigma_Y^2)$.

(vi) Calculate the minimum sample size to give a 95% confidence interval for the expectation of Y with a width of less than 5 mph, assuming that the variance of Y is known and equal to $\sigma^2 = 4.8^2$. [3]

The analyst suggests that the variance of Y can be estimated instead of assuming it is known. However, the insurer's management, who wish to control costs, have responded to say this would increase the sample size required to meet the target confidence interval.

(vii) Comment on the validity of the management's response. [2]

[Total 12]

8 A teacher runs a summer school and is trying to justify the cost to their local council.

To support their justification, the teacher tested ten students before and after attending summer school, and obtained the scores shown below (x and y, respectively):

Student (i)	1	2	3	4	5	6	7	8	9	10
Exam score before (x_i)	63	41	45	63	53	55	56	49	61	32
Exam score after (y_i)	70	52	51	66	58	61	61	54	65	45

For these data:

$$\sum x_i = 518 \qquad \sum y_i = 583 \qquad \sum x_i^2 = 27{,}760 \qquad \sum y_i^2 = 34{,}533 \qquad \sum x_i y_i = 30{,}891$$

(i) Fit a linear regression of y on x. [4]

(ii) Calculate Pearson's correlation coefficient between the two sets of exam scores. [1]

(iii) Comment on the relationship between the two sets of exam scores, based on your answer to part (ii). [2]

(iv) Calculate the proportion of the variance that can be explained by the linear model in part (i). [1]

A student has scored 45 in the exam before attending the summer school.

(v) Construct a 95% prediction interval for this student's score after attending summer school. [6]

[Total 14]

9 A large insurance company sells policies that pay out in the event of a cyber attack. The insurance claims (in $) received at their claims centre over the last year have been analysed. The results of this analysis are as follows:

Size of claim, S	$0 \leq S < 250$	$250 \leq S < 500$	$500 \leq S < 750$	$750 \leq S < 1{,}000$	Over 1,000
Number of claims	102	59	34	23	9

(i) Identify which one of the following data labels is appropriate for the above analysis:

A cross-sectional data
B truncated data
C longitudinal data
D censored data.

[1]

(ii) You are given that the maximum claim size is $5,000.

(a) Determine the sample mean of the data, stating any assumptions that you make. [2]

(b) Determine the expected claim size in each of the five ranges for S in the analysis above, if an exponential distribution with parameter λ is used as the distribution of the claim sizes. [4]

[**Hint:** The value of λ can be estimated using the method of moments.]

(c) Test at the 1% level if an exponential distribution with parameter λ is an appropriate distribution for the claim sizes. [4]

An actuary is investigating if sizes of claims vary according to the region in which the claimant is located. The data is split into four different regions. The results of these are given below:

Region 1:

Size of claim, S	$0 \leq S < 250$	$250 \leq S < 500$	$500 \leq S < 750$	$750 \leq S < 1{,}000$	Over 1,000
Number of claims	29	11	9	5	3

Region 2:

Size of claim, S	$0 \leq S < 250$	$250 \leq S < 500$	$500 \leq S < 750$	$750 \leq S < 1{,}000$	Over 1,000
Number of claims	31	18	6	4	1

Region 3:

Size of claim, S	$0 \leq S < 250$	$250 \leq S < 500$	$500 \leq S < 750$	$750 \leq S < 1{,}000$	Over 1,000
Number of claims	17	14	8	6	2

Region 4:

Size of claim, S	$0 \leq S < 250$	$250 \leq S < 500$	$500 \leq S < 750$	$750 \leq S < 1{,}000$	Over 1,000
Number of claims	25	16	11	8	3

(iii) (a) Determine the expected claim sizes if claim size is independent of region. [2]

(b) Test at the 1% level if claim size is independent of region. [8]

[Total 21]

END OF PAPER